The Integrated K3 Transformer

Deep Dive into Kimi K3 — Chapter 10

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Foundation Books Project

The one idea of this chapter

One stream, three mixers, four state lifetimes:
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Goal: fix coordinates, type one decoder layer, assemble the 93-layer backbone, trace a multimodal prefill and an accepted decode exactly, and keep every ledger, precision island, and claim in scope.

Roadmap

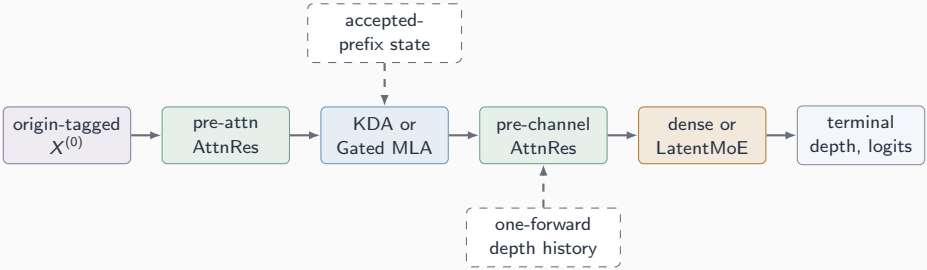
One ordered transition

Prefill and accepted decode

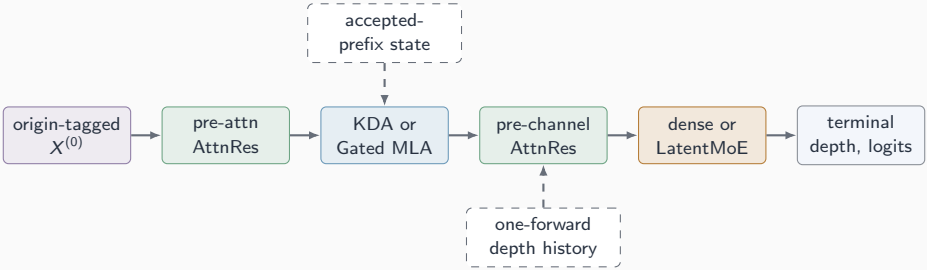
Ledgers, precision, and diagnosis

One ordered transition

Two retrievals, two state edges, one schedule

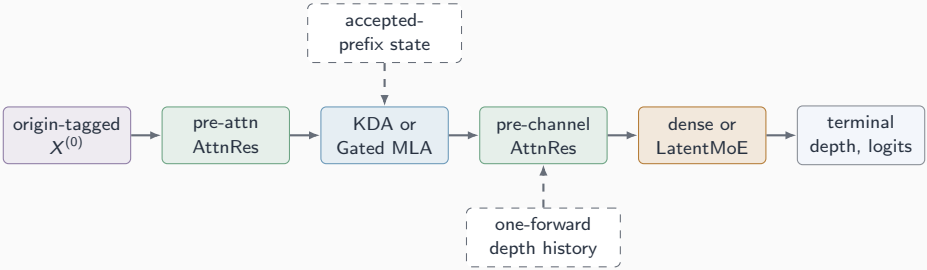


Two retrievals, two state edges, one schedule



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- Production: $(K, K, K, G)^{23}, G$ — 69 KDA, 24 global; dense first channel, then 92 routed; 186 depth points in groups $7 \times 12 + 9$, nine terminal sources. [REPORTED: K3 Table 1; snapshot c5d1dd4c]

Prefill and accepted decode

An exact trace, then one accepted token

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- Continuation from Σ_{10} matches an eleven-row full forward on all 12 logits exactly; finite precision needs a declared tolerance.

Ledgers, precision, and diagnosis

Ledgers, precision, and evidence

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Order and state define the computation. Next: Chapter 11's Advanced Reading on MoBA and learned block-sparse attention.